

Chapter 15

Seaweeds in the Antarctic Marine Coastal Food Web



Fernando R. Momo, Georgina Cordone, Tomás I. Marina, Vanesa Salinas, Gabriela L. Campana, Mariano A. Valli, Santiago R. Doyle, and Leonardo A. Saravia

Abstract Antarctic macroalgae are the basis of marine food webs in most coastal environments, especially the more confined ones such as bays and fjords. Whether through direct consumption or via detritus, their role in maintaining biodiversity is

F. R. Momo (✉)

Instituto de Ciencias, Universidad Nacional de General Sarmiento,
Los Polvorines, Buenos Aires, Argentina

Departamento de Ciencias Básicas, Universidad Nacional de Luján (UNLu), Luján, Buenos Aires, Argentina

e-mail: fmomo@campus.ungs.edu.ar

G. Cordone

Centro Nacional Patagónico (CCT CONICET-CENPAT), Centro para el Estudio de Sistemas Marinos (CESIMAR), Puerto Madryn, Chubut, Argentina

e-mail: gcordone@cenpat-conicet.gob.ar

T. I. Marina

Consejo Nacional de Investigaciones Científicas y Técnicas (CONICET), Centro Austral de Investigaciones Científicas (CADIC), Ushuaia, Tierra del Fuego, Argentina

e-mail: tomasimarina@cadic-conicet.gob.ar

V. Salinas · L. A. Saravia

Instituto de Ciencias, Universidad Nacional de General Sarmiento,
Los Polvorines, Buenos Aires, Argentina

e-mail: vsalinas@campus.ungs.edu.ar; lsaravia@campus.ungs.edu.ar

G. L. Campana

Departamento de Biología Costera, Instituto Antártico Argentino, Buenos Aires, Argentina

Departamento de Ciencias Básicas, Universidad Nacional de Luján (UNLu),
Luján, Buenos Aires, Argentina

e-mail: gcampana@dna.gov.ar

M. A. Valli

Departamento de Ciencias Básicas, Universidad Nacional de Luján (UNLu),
Luján, Buenos Aires, Argentina

S. R. Doyle

Instituto de Ecología y Desarrollo Sustentable (INEDES), Universidad Nacional de Luján (UNLu), Luján, Buenos Aires, Argentina

e-mail: sdoyle@campus.ungs.edu.ar

essential. However, their relevance is due not only by direct trophic interactions but also by indirect feedbacks, since macroalgae act as a habitat and refuge for multiple benthic organisms and as a substrate for epiphytic microalgae. Macroalgae also establish relations of exploitative competition, apparent competition, and mutualism. The control over the biomass and diversity of the macroalgae itself does not seem to be due to trophic interactions (top-down control) but rather to competition and diverse abiotic factors such as substrata and light availability or physical disturbances (ice scouring). The extreme connectivity of trophic networks linked to algae and their detritus determines that food webs are robust to local extinctions; however, non-trophic interactions indicate that changes that affect the growth, biomass, and distribution of macroalgae can have dramatic effects on the diversity of their associated fauna and, indirectly, on the networks of consumers of that fauna. In this chapter, we present a detailed description of macroalgae relationship networks and analyze the stability of the Antarctic community using food web theory.

Keywords Biological interactions · Community structure · Ecological networks · Trophic level

15.1 Introduction

Macroalgae and their detritus or fragments are the basal energy source of Antarctic coastal food webs. Approximately 150 species of macroalgae have been reported at Antarctica (Wiencke et al. 2014; Pellizzari et al. 2017; see also Chap. 2 by Oliveira et al.). Several local studies proved that macroalgae together with microphytobenthos constitute a direct pathway of energy and matter into organisms that feed on them (Wiencke et al. 2014; Ahn et al. 2016) and indirectly through detritus pathway (Iken et al. 1997; Tatián et al. 2004; Amsler et al. 2005, 2012; Quartino et al. 2008; Campana et al. 2018). Besides this, benthic algae in some locations can have high primary productivity similar to or higher than phytoplankton production (Fogg 1977). Furthermore, Antarctic macroalgae constitute structural and even chemical refuges from predation (Amsler et al. 1999, 2009), habitat for a variety of associated fauna (Huang et al. 2007) that provide a large fraction of the secondary production to the benthos (Gómez et al. 2009), and substrate for epiphytic communities (Majewska et al. 2016). Thus, macroalgae play a fundamental role in the food web being able to influence ecosystem dynamics and stability through propagation of direct and indirect effects.

The network of interspecific relationships involving seaweeds is very complex and includes not only consumption of algae but also several types of relations. Direct interactions are, for instance, competition among different macroalgae species for space (Smale and Barnes 2008; Quartino et al. 2013; Campana et al. 2009,

2018), when hard substrate is a limiting resource, or for light (Klöser et al. 1994, 1996; Deregibus et al. 2016). On the other hand, seaweeds can establish relationships with potentially harmful light-blocking epiphytes, whereas macroalgae provide a substrate for these organisms; however, seaweeds are chemically defended against mesograzers (and other herbivores; see Chap. 18 by Amsler et al.) that, in turn, use them as a refuge from predation: the macroalgae benefit in return because the mesograzers remove epiphytic algae (Amsler et al. 2014), and in this way, commensalism can become mutualism.

However, not only direct effects play an important role: several seaweed species share predators and establish a complex dynamics that may be observed as much as apparent competition or “apparent” mutualism depending on the density of predators and the closeness of macroalgae species involved. When a group of species establish different types of interactions (i.e., trophic, commensalism, mutualism) in an intricate manner, it is challenging to decide if the regulation of numbers or biomass is mainly controlled by top-down, bottom-up, or wasp-waist effects or by different and nontrivial combinations of them. In consequence, it is necessary to take into account the complexity of the ecosystem and all its ecological interactions (trophic and non-trophic) to understand its stability in response to environmental changes. In particular, macroalgal species are exposed to different environmental disturbances such as sediment input or ice scouring (Sahade et al. 2015; Quartino et al. 2013; Deregibus et al. 2016), resulting in a complex mosaic of effects. In this sense, the understanding of the effects of climate change in the western Antarctic Peninsula and a possible losses and gains in biodiversity urgently needs a deep knowledge of the relations between species’ functional roles and the ecosystem structure (Woodward et al. 2010).

15.2 Food Webs and Seaweeds

The complex network of interactions varies in space and time, and not all interactions take place simultaneously (Poisot et al. 2015), but the interactions should not be studied separately but be understood as a whole. For this purpose, we have a powerful tool: the food web theory (Delmas et al. 2018).

After comparing 19 food web properties, Dunne et al. (2004) concluded that the excessively low percentage of basal taxa in marine food webs compared to other systems is clearly an artifact due to the poor resolution of primary producers and consumer links to them. One of the methodological strengths of the food web studied here is the high taxonomic resolution of basal nodes. A good taxonomic resolution of the lower trophic levels, such as the macroalgal community, is essential to understand ecosystem functioning, since there seems to be a species-specific selective consumption (Iken et al. 1997). Implications for ecosystem functioning and stability are only possible to elucidate in food webs where the species involved in energy and matter transfer processes are well represented.

A first step to identify the seaweeds embedded in the functional context of their community is to visualize the food web in which they are integrated. For example, Fig. 15.1 shows the food web of the Potter Cove ecosystem (Marina et al. 2018). We can see that the base of the pyramid is filled with several macroalgal species accompanied by fresh and old detritus and microalgae. Arrows indicate matter and energy fluxes through predation; the node size is proportional to its number of connections; the placement of the node on the vertical coordinate is related to its weighted trophic level. Also, macroalgae contribute to the detritivore pathway by decomposition and accumulation in the cove, generating a link between macroalgae species and detritus that is not represented in Fig. 15.1. However, we consider it in our studies and conclusions.

An understanding of the relations between species functional roles and ecosystem structure is an indispensable step toward the comprehension of change in the western Antarctic Peninsula and subsequent biodiversity loss and gain (Woodward et al. 2010). Several network properties are commonly used to describe food webs (Dunne et al. 2002): (1) number of species, S ; (2) total number of interactions or trophic links, L ; (3) number of interactions per species or linkage density, L/S ; (4) connectance or trophic links divided by total number of possible interactions, $C = L/S^2$; (5) percentage of top species (species with preys, but without predators); (6) intermediate species (species having preys and predators); (7) basal species (species with predators/consumers, but without preys); and (8) percentage of omnivores (species eating prey from more than one trophic level). Additionally, the topology of the food web can be studied by measuring three more properties: (9) characteristic path length (ChPath), which is the average shortest path length between all pairs of species; (10) clustering coefficient (CC), which is the average fraction of pairs of species one link away from a species that are also linked to each other; and (11) distribution degree, which is the fraction of trophic species $P(k)$ that have k or more trophic links (both predator and prey links) (Albert and Barabási 2002). These last three metrics give us information related to the degree of self-organization shown by the web.

The trophic levels (TL) of species were calculated using the short-weighted TL (Williams and Martinez 2004). Short-weighted trophic level is defined as the average of the shortest TL and prey-averaged TL . Shortest TL of a consumer in a food web is equal to 1+ the shortest chain length from this consumer to any basal species. Prey-averaged TL is equal to 1+ the mean TL of all consumer's trophic resources, calculated as follows:

$$TL_j = 1 + \sum_{i=1}^S l_{ij} \frac{TL_i}{n_j}$$

where TL_j is the trophic level of species j ; S is the total number of species in the food web; l_{ij} are the elements of the connection matrix with S rows and S columns; for column j and row i , l_{ij} is 1 if species j consumes species i and 0 if not; and n_j is the number of prey species in the diet of species j . Therefore, short-weighted TL yields a minimum estimate of TL and assumes a value of 1.0 for basal species (Williams and Martinez 2004). We considered the mean TL of the web as the average of all species' TL .

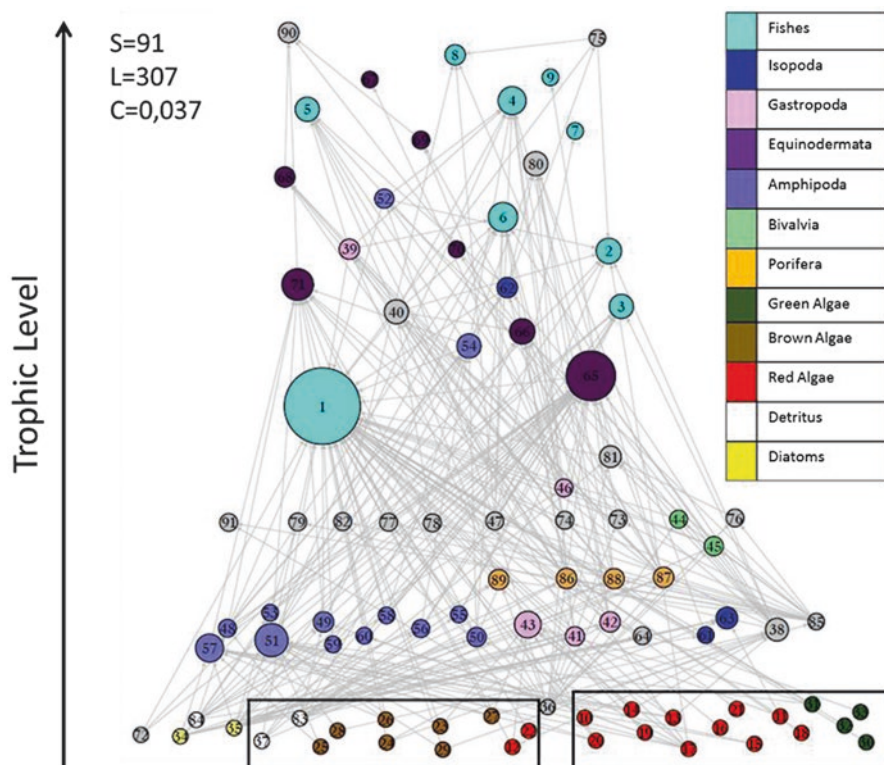


Fig. 15.1 Food web of Potter Cove. Vertical position is related to trophic level. Node size is proportional to the total degree (in and out). Node colors are by functional group. Nodes inside the boxes are the basal species. Network was plotted with Visone (version 2.9.2). ¹*Notothenia coriiceps*, ²*Notothenia rossii*, ³*Lepidonotothen nudifrons*, ⁴*Trematomus newnesi*, ⁵*Trematomus bernacchii*, ⁶*Harpagifer antarcticus*, ⁷*Parachaenichthys charcoti*, ⁸*Chaenocephalus aceratus*, ⁹*Protomyctophum* sp., ¹⁰*Callophyllis atrosanguinea*, ¹¹*Curdiea racovitzae*, ¹²*Georgiella confluens*, ¹³*Gigartina skottsbergii*, ¹⁴*Iridaea cordata*, ¹⁵*Myriogramme manginii*, ¹⁶*Neuroglossum delessieriae*, ¹⁷*Palmaria decipiens*, ¹⁸*Pantoneura plocamioides*, ¹⁹*Picconiella plumosa*, ²⁰*Plocamium cartilagineum*, ²¹*Porphyra plocamiestris*, ²²*Trematocarpus antarcticus*, ²³*Adenocystis utricularis*, ²⁴*Ascoseira mirabilis*, ²⁵*Desmarestia anceps*, ²⁶*Desmarestia antarctica*, ²⁷*Desmarestia menziesii*, ²⁸*Geminocarpus geminatus*, ²⁹*Phaeurus antarcticus*, ³⁰*Lambia antarctica*, ³¹*Monostroma hartiottii*, ³²*Urospora penicilliformis*, ³³*Ulothrix* sp., ³⁴Epiphytic diatoms, ³⁵Benthic diatoms, ³⁶Phytoplankton, ³⁷Aged detritus, ³⁸Nereidae, ⁴⁰*Austrodoris kerguelensis*, ⁴¹*Eatoniella* sp., ⁴²*Nacella concinna*, ⁴³*Laevilacunaria antarctica*, ⁴⁴*Dacrydium* sp., ⁴⁵*Laternula elliptica*, ⁴⁶*Neobuccinum eatoni*, ⁴⁷*Euphausia superba*, ⁴⁸*Paradexamine* sp., ⁴⁹*Eurymera monticulosa*, ⁵⁰*Pontogeneiella* sp., ⁵¹*Gondogeneia antarctica*, ⁵²Hyperiid, ⁵³*Pariphimedia integricauda*, ⁵⁴*Bovallia gigantea*, ⁵⁵*Cheirimedon femoratus*, ⁵⁶*Gitanopsis antarctica*, ⁵⁷*Protebbbingia gracilis*, ⁵⁸*Waldeckia obesa*, ⁵⁹*Hippo-Orcho* (*Hippomedon kergueleni* and *Orchomene plebs* collapsed), ⁶⁰*Oradarea bidentata*, ⁶¹*Serolis* sp., ⁶²*Glyptonotus antarcticus*, ⁶³*Plakarthrium puncattissimum*, ⁶⁴*Hemiarthrum setulosum*, ⁶⁵*Ophionotus victoriae*, ⁶⁶*Odontaster validus*, ⁶⁷*Diplasterias brucei*, ⁶⁸*Odontaster meridionalis*, ⁶⁹*Perknaster fuscus* antarcticus, ⁷⁰*Perknaster aurorae*, ⁷¹*Sterechinus neumayeri*, ⁷²squids, ⁷³Copepods, ⁷⁴Ascidians, ⁷⁵*Octopus* sp., ⁷⁶Oligochaetes, ⁷⁷Hydrozoa, ⁷⁸Bryozoa, ⁷⁹Priapulids, ⁸⁰*Parborlasia corrugatus*, ⁸¹Salpidae, ⁸²Mysida, ⁸³fresh detritus, ⁸⁴necromass, ⁸⁵zooplankton, ⁸⁶*Haliconidae* sp., ⁸⁷*Stylo-Myca* (*Stylocordila borealis* and *Mycale acerata* collapsed), ⁸⁸*Rosella* sp., ⁸⁹*Dendrilla antarctica*, ⁹⁰*Urticinopsis antarctica*, ⁹¹*Malacobelemn non daytoni*. (Modified from Marina et al. 2018)

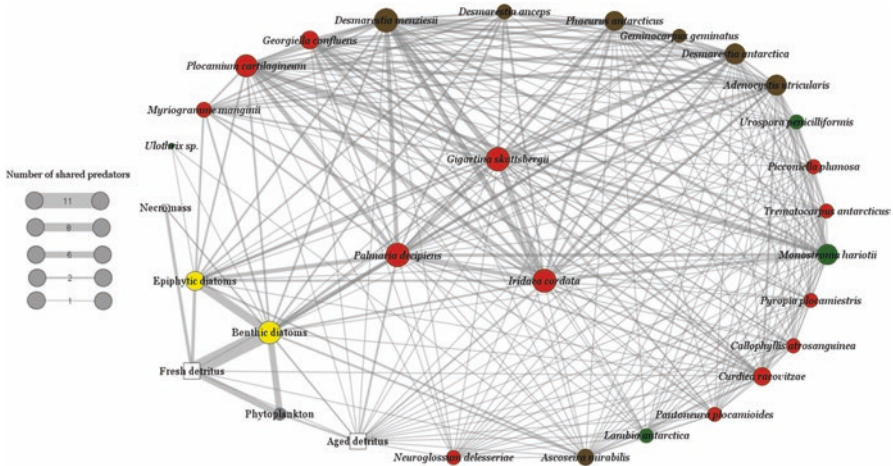


Fig. 15.2 Common-enemy graph for basal species of Potter Cove food web. Nodes represent species and link indirect interactions (presence of shared predators). Only prey species are shown. Node colors identify different preys (type of algae, or detritus, or microalgae). Node and link width are proportional to the number of shared predators. (Modified from Marina et al. (2018) and Cordone et al. (2018))

The influence of the seaweed community in Potter Cove ecosystem is notorious not only in the structure of the food web but also in the functioning. For instance, the maximum trophic level for Potter Cove food web is 4.27 (Marina et al. 2018), lower than most other food webs studied (Dunne et al. 2002, 2004). This implies that top and basal species (dominated by macroalgae) are separated by few intermediate taxa. Therefore, the transfers of energy or nutrients from the base to the top of Potter Cove food web is small, so that the number of times chemical energy is transformed from a consumer's diet into a consumer's biomass along the food web is also small. The mean trophic level for the mentioned food web is also low (2.10), which is a consequence of the fact that most predators at intermediate levels (e.g., amphipods, isopods, bivalves, *Notothenia coriiceps*) feed predominantly on algae species and/or detritus, being mainly the product of dead and decomposed macroalgae in Potter Cove (Iken et al. 1998; Quartino et al. 2008; see also Chap. 8 by Quartino et al.). The macroalgal detritus decomposes and is eaten by detritivores and suspensivores (e.g., sponges, ascidians, bryozoans, cnidarians), supporting an important amount of the secondary production (Tatián et al. 2004). The obtained low mean trophic level for Potter Cove food web clearly shows what species-specific and/or community studies have suggested. These characteristics of ecological communities have a high impact on ecosystem functioning, such as nutrient and carbon cycling, and trophic cascades (Post 2002).

However, our interest here is focused on macroalgae. To explore mainly their interactions, we can use another useful information due to the secondary graph of common-enemy graph (Fig. 15.2). This graph represents prey species linked with each other according the existence of shared predators. In the representation of

Fig. 15.2, the thickness of each link joining a couple of nodes is proportional to the number of enemies shared by these nodes. The common-enemy graph is hyperconnected having a high-density value of interactions per node (Cordone et al. 2018); this is an indicator of the existence of multiple alternative energy paths. This hyperconnectivity makes the prediction of indirect relationships between species very difficult, as each interaction involves positive and negative effects between species abundances (Holt and Lawton 1994).

15.3 Network Dynamics and Robustness

An immediate question that we can try to answer is the following: can we predict the behavior of the community against the (local) extinction of some algae? This important question cannot be easily answered by field observations or experiments; however, we can carry out *in silico* experiments using our food web (Cordone et al. 2018). We made a virtual experiment trying to determinate if successive extinctions of base species in Potter Cove could produce phase transitions in the emergent properties of the food web. With this objective, we simulated extinctions by deleting nodes (seaweeds, detritus, microalgae) according to different sequences established by the degree of each macroalgae (total number of trophic interactions per node) and their biomass in Potter Cove. Four sequences of extinctions were used: random order, degree in ascending and descending order, and biomass in ascending order.

The method quantifies secondary extinctions taking into account the existence of different extinction thresholds (Schleuning et al. 2016). That means that a species suffers a secondary extinction when it loses a given percentage of its preys (Solé and Montoya 2001). We define the threshold (v) as the minimum level of energy necessary for species' survival. After each node removal, the fraction of original incoming energy $e(i)$ is calculated for each species i , and when this fraction is equal or less than the threshold, the species is secondarily lost. The classical topological approach assumes that v is equal to 0, so only when the energy inflow is null a species goes extinct (Bellingeri and Bodini 2013).

The results of the *in silico* experiment considering different thresholds for extinction and the four sequences of primary elimination are indicated in Fig. 15.3. We found that the Potter Cove food web is relatively stable to macroalgae loss, but a significant number of secondary extinctions were obtained beyond a 50% threshold (Cordone et al. 2018). The elimination of macroalgae species from the Potter Cove food web does not seem to generate a catastrophic cascade of secondary extinctions, suggesting that the Potter Cove food web could be more robust than other similar networks (Allesina et al. 2006). Most connected macroalgae shared more predators, which could indicate that these macroalgae species are functionally redundant (i.e., species with equivalent trophic interactions). Functional redundancy has important consequences in potential cascade extinctions, since it increases food web resistance by means of availability of alternative preys (Borrvall et al. 2000; Petchey et al. 2008). However, these results are tied to the limitations

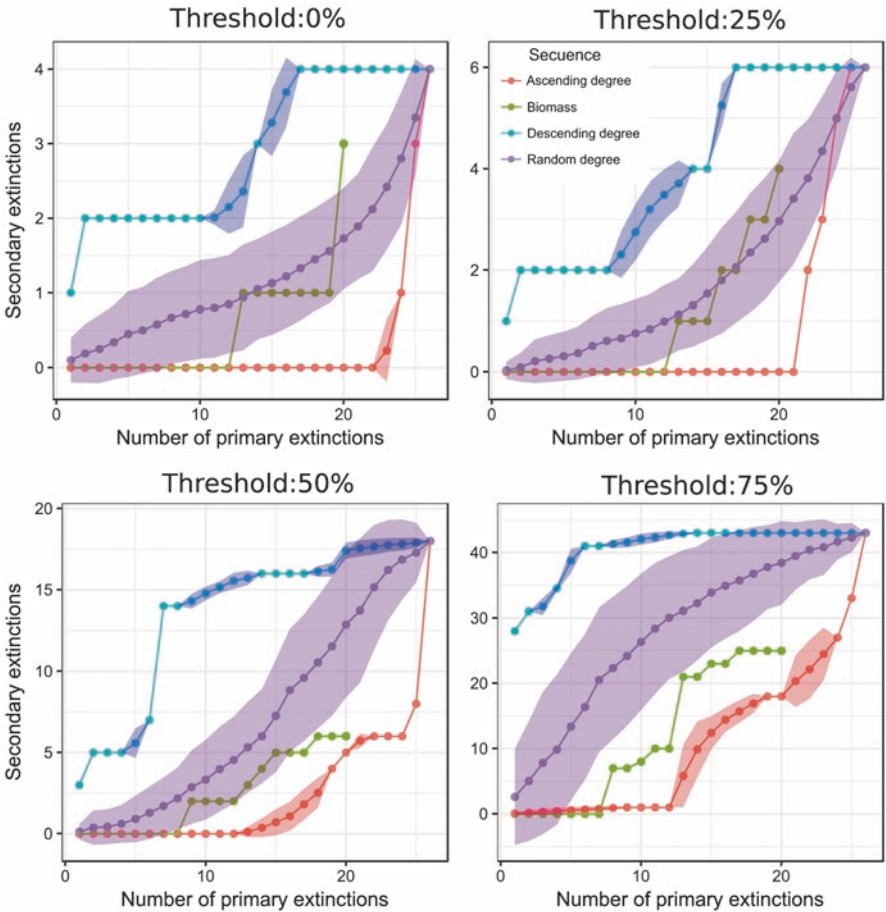


Fig. 15.3 Secondary extinctions vs extinctions (total, primary plus secondary) when macroalgae species and detritus are removed. The secondary extinctions after primary extinctions take place or not depending on the numbers of preys that the predator needs to survive. This number is determined by the threshold of extinctions as the number of original preys multiplied by threshold. Each box corresponds to a simulation in which the threshold of extinctions was constant and different (0%, 25%, 50%, and 75%). The macroalgae loss was made following the four different sequences: red, ascending degree; blue, descending degree; violet, random degree – mean and interval confidence; and green, biomass – in ascending order

of our method. Typically, the topological approach underestimates the actual number of secondary extinctions and in consequence overestimates food web robustness. The introduction of the threshold effect gives us a more accurate prediction (Fig. 15.4).

Despite this, the topological approach enables the analysis of complex food webs, since it only requires knowledge of network structure and could be used as a proxy of food web stability (Eklöf et al. 2013). Furthermore, we have to

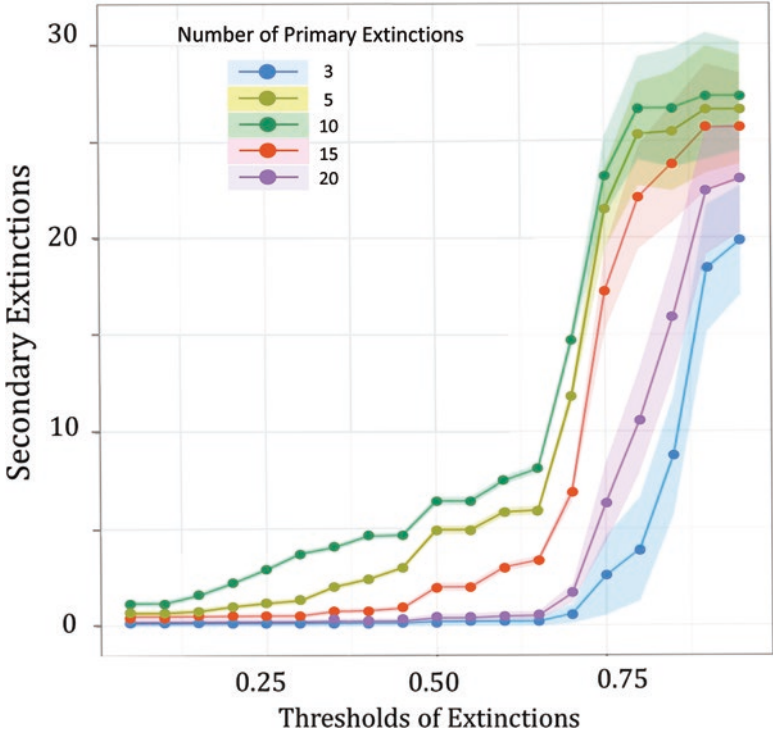


Fig. 15.4 This figure summarizes the effect of varying thresholds in the number of secondary extinctions recorded. We tested thresholds from 5% to 95% by 5% when macroalgae are primary lost (blue, 3; brown, 5; green, 10; red, 15; pink, 20). Points are means of secondary extinctions when x number of primary extinctions took place at a particular threshold (by random simulations). Shaded area represents confidence interval of the series. Secondary extinctions recorded at a fixed threshold (0.25, 0.50, 0.75) from Fig. 15.3. (Modified from Cordone et al. 2018)

consider that this food web includes all recorded trophic interactions, though these interactions do not always occur simultaneously in time and space. We cannot ignore the potentially confounding effects of seasonality and spatial sampling (Ings et al. 2009). In fact, the network is an average representation across seasons and different habitats, and dynamic stability depends more on how interactions are materialized in time and space. Some nodes are pulsatile, such as the massive influx of Krill (Fuentes et al. 2016), which is difficult to reflect in a static description of the network topology. The consideration of all these factors might change our estimation of the fragility of Potter Cove food web. Our current efforts are focused on incorporating spatial and temporal variability into the modeling of Potter Cove food web to achieve a more realistic picture of this ecosystem and its macroalgae species.

15.4 Non-Trophic Interactions

Ecological systems are complex and multifaceted in the nature of their interactions and should be defined not only by lists of co-occurring species but also by the variety of interactions that take place between them. In order to satisfy different requirements, species interact in many ways with multiple partners and make associations that imply more than a trophic relationship. Great advances and multiple practical applications have been developed since the study of the structure, intensity, and dynamics of trophic interactions. However, ecological interactions between co-existing species involve much more than simply feeding (Bascompte et al. 2003; Kéfi et al. 2012; Pocock et al. 2012; Kéfi et al. 2015, 2016a).

Decades of empirical and theoretical studies have shown that specific non-trophic interactions, such as habitat modification, stress minimization, ecosystem engineering, and behavioral changes, can play important roles for community structure and ecosystem functioning (Pocock et al. 2012; Kéfi et al. 2015, 2016a). Some works have recognized this importance and have focused their study on isolated networks of non-trophic interactions such as mutualistic networks (Jordano et al. 2009) and, more recently, have incorporated different types of interaction (e.g., mutualistic, competition, and trophic) in a single network usually named “multiplex” (Kéfi et al. 2012, 2015, 2016b; Pocock et al. 2012; García-Callejas et al. 2018).

To know how important non-trophic interactions are for the ecosystem, functioning triggers many interesting questions around these relationship types: How many direct non-trophic interactions are there and how are they distributed among species? Do the topological properties of trophic and non-trophic networks differ? Can simple species attributes help predict the type of interaction between two species? Can trophic characteristics help predict the non-trophic webs (Kéfi et al. 2015)? Here, we have focused on the first question to describe non-trophic interactions that involve macroalgae species from the Potter Cove food web.

The number of non-trophic interactions identified in Potter Cove marine ecosystem duplicates the number of trophic relationships: 1091 versus 454, respectively. Here, non-trophic interactions include competition or negative interactions (−/−) considered when a pair of species share at least one prey and represent the 74% of the total number of non-trophic interactions; mutualism or positive interactions that involve a benefit of both interacting species (+/+) that represent the 13.6% of the total number of non-trophic interactions; and commensalism (+/0) and amensalism (−/0) or neutral interactions in which one species benefit or not-benefit, respectively, and the other species is not affected by the link, and these interactions represent the 12.4% of the total number of non-trophic interactions. The distribution of these interactions (Fig. 15.5) depends on the trophic classification of the species in basal, intermediate, or top species, e.g., competition, as we defined here, occurs between intermediate and/or top species, which means that most non-trophic interactions in Potter Cove ecosystem occur between species of high trophic levels. So, we should wonder what the role of macroalgae in the non-trophic relationships is.

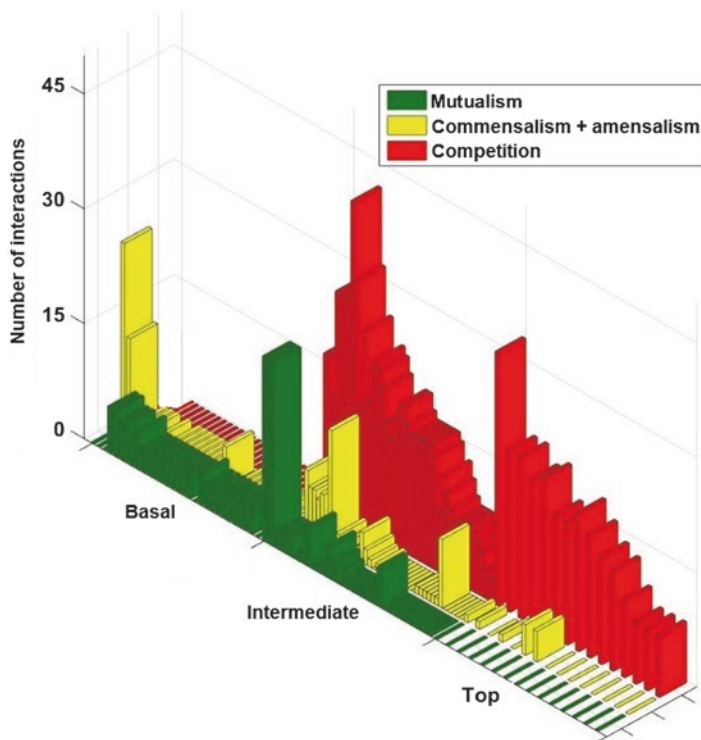


Fig. 15.5 Number of non-trophic interactions for basal, intermediate, and top species. Each color represents a type of interaction

Many works highlight the importance of studying the structure of non-trophic networks versus the trophic network by suggesting that, in general, trophic networks tend to be compartmentalized, while positive non-trophic networks are more nested (Bascompte et al. 2003; Thébault and Fontaine 2010), which minimizes competition, increases the number of co-existing species, and makes the community more robust against random extinctions and diversity loss (Memmott et al. 2004; Fortuna and Bascompte 2006).

In food webs, it is argued that modular patterns tend to increase the stability of the network by retaining the impacts of disturbances within each module and minimizing that impact on other modules in the network. In the particular case of Potter Cove, it was observed that the food web has not modular characteristics, reason to suppose that the network should be not very robust against random disturbances. However, it has been shown that the trophic network is highly robust against different types of disturbances, which makes it possible to assume that positive non-trophic interactions could have a stabilizing role in the Potter Cove ecosystem. Then, we should not fail to recognize that macroalgae are a key component of the system because they represent, directly or indirectly, a stabilizing species.

15.5 Final Remarks

Network science offers a great opportunity to understand species in their context and their relation to the community and the ecosystem. In our studies, the incorporation of a network perspective shows the relevance of macroalgae species to the entire ecosystem of Potter Cove. We observed that Potter Cove food web is relatively robust to local extinctions of macroalgal species under a topological approach (considering extinction thresholds), but we also observed that a network collapse could be reached by increasing the extinction threshold. This result demonstrates that effects on single species could propagate to the entire network. Potter Cove ecosystem is experiencing rapid climate-related changes in environmental factors that have affected the benthic system, including the macroalgal community (Schloss et al. 2012; Quartino et al. 2013; Sahade et al. 2015; Deregibus et al. 2016; Campana et al. 2018). In this sense, species-specific responses of macroalgae to changing environmental factors should be explored as negative effects at the species level could propagate through the food web, leading to changes in the structure and function of this system.

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